

Chomsky Normal Form (CNF)

When working with CFGs, it is often convenient to have them in simplified form.

This is useful, when we have algorithms working on CFGs.

Definition

Chomsky Normal Form (CNF)

A context free grammar is in CNF if every rule is in the form of:

$$A \rightarrow BC$$

$$A \rightarrow a$$

a is a terminal, and A, B, C are any variable except that B & C may not be start variable.

We also allow $S \rightarrow \epsilon$ for the start variable
(only for the start variable)

Theorem 2.9

Any context free language is generated by a context-free grammar in chomsky Normal form.

Proof: (By construction)

Instead of going through detailed proof, we will go through an example to understand this.

$$S \rightarrow ASA \mid aB$$

$$A \rightarrow B \mid S$$

$$B \rightarrow b \mid \epsilon$$

Let's convert this to CNF

1. we add a new starting variable.

(This guarantees that the start variable does not appear on right hand side of any rule.

ex: $S_0 \rightarrow S$

2. we take care of ϵ -rules.

we remove $A \rightarrow \epsilon$ rules where A is not the start state.

for each occurrence of A on the right hand side of a rule, we add a new rule with that occurrence deleted.

want to remove $A \rightarrow \epsilon$

Ex: $R \rightarrow uA \vee AW$

* $R \rightarrow uA \vee AW \mid u \vee AW \mid uA \vee W \mid u \vee W$

$R \rightarrow A \Rightarrow R \rightarrow \epsilon$ (only if we did not remove $R \rightarrow \epsilon$)

3. Handle all unit rules

- (i) we remove unit rule $A \rightarrow B$
- (ii) when we see $B \rightarrow u$, we add $A \rightarrow u$ unless we have already removed it earlier
 - * u is a string of variables and terminals

4. Convert all remaining rules into proper form

- (i) we replace each rule in the form of

$A \rightarrow u_1 u_2 \dots u_k$, where $k \geq 3$, and u_i is a variable or a terminal symbol, with rules

$$A \rightarrow u_1 A_1$$

$$A_1 \rightarrow u_2 A_2$$

$$A_2 \rightarrow u_3 A_3 \dots \dots A_{k-2} \rightarrow u_{k-1} u_k$$

we replace any u_i in the preceding rule(s) with the variable u_i and

$$u_i \rightarrow u_i$$

$$A \rightarrow u_1 u_2 u_3 \dots u_k$$

$$A \rightarrow u_1 A_1$$

$$A_1 \rightarrow u_2 u_3 \dots u_k$$

$$A \rightarrow u_1 A_1$$

$$A_1 \rightarrow u_2 A_2$$

$$A_2 \rightarrow u_3 u_4 \dots u_k$$

⋮

Let's do an example

$$S \longrightarrow ASA \mid aB$$

$$A \longrightarrow B \mid S$$

$$B \longrightarrow b \mid \epsilon$$

$$S_0 \longrightarrow S$$

$$S \longrightarrow ASA \mid aB$$

$$A \longrightarrow B \mid S$$

$$B \longrightarrow b \mid \epsilon$$

Step 01

Remove $B \rightarrow \epsilon$

$$S_0 \longrightarrow S$$

$$S \longrightarrow ASA \mid aB \mid a$$

$$A \longrightarrow B \mid \epsilon \mid S$$

$$B \longrightarrow b$$

$$S_0 \longrightarrow S$$

remove $A \rightarrow \epsilon$

$$S \longrightarrow ASA \mid SA \mid AS \mid S \mid aB \mid a$$

$$A \longrightarrow \cancel{B} \mid S$$

$$B \longrightarrow b$$

$$S_0 \rightarrow S$$

remove $A \rightarrow B$

$$S \rightarrow ASA | SA | AS | S | aB | a$$

$$A \rightarrow b | \cancel{S}$$

$$B \rightarrow b$$

$$S_0 \rightarrow S$$

remove $A \rightarrow S$

$$S \rightarrow ASA | SA | \cancel{AS} | aB | a$$

$$A \rightarrow b | ASA | SA | AS | aB | a$$

$$B \rightarrow b$$

$$S_0 \rightarrow \cancel{S}$$

remove $S \rightarrow S$

$$S \rightarrow ASA | SA | AS | aB | a$$

$$A \rightarrow b | ASA | SA | AS | aB | a$$

$$B \rightarrow b$$

$$S_0 \rightarrow ASA | SA | AS | aB | a$$

remove $S_0 \rightarrow S$

$$S \rightarrow ASA | SA | AS | aB | a$$

$$A \rightarrow b | ASA | SA | AS | aB | a$$

$$B \rightarrow b$$

$$S_0 \rightarrow AU_1 | SA | AS | aB | a$$

$$\text{Handle } S_0 \rightarrow \underline{ASA}$$

$$U_1 \rightarrow SA$$

$$U_1 \rightarrow SA$$

$$S \rightarrow ASA | SA | AS | aB | a$$

$$S_0 \rightarrow AU_1$$

$$A \rightarrow b | ASA | SA | AS | aB | a$$

$$B \rightarrow b$$

$$S_0 \rightarrow AU_1 | SA | AS | U_2B | a$$

$$U_1 \rightarrow SA$$

$$U_2 \rightarrow a$$

$$S \rightarrow ASA | SA | AS | aB | a$$

$$A \rightarrow b | ASA | SA | AS | aB | a$$

$$B \rightarrow b$$

$$S_0 \rightarrow AU_1 | SA | AS | U_2B | a$$

$$U_1 \rightarrow SA$$

$$U_2 \rightarrow a$$

$$S \rightarrow AU_1 | SA | AS | U_2B | a$$

$$A \rightarrow b | AU_1 | SA | AS | U_2B | a$$

$$B \rightarrow b$$